

Week 2

DevOps Internship

Release Date: 27 July 2026

This week, you'll learn **Git, GitHub & DevOps Collaboration**. Mastering version control helps teams collaborate efficiently, track code changes, and streamline modern DevOps workflows.

Topic:- Git, GitHub & DevOps Collaboration

This week, you are expected to learn:

- Introduction to Version Control
- Git Installation & Configuration
- Git Workflow
- GitHub Fundamentals
- Creating & Cloning Repositories
- Staging, Commit & Push
- Branching & Merging
- Git Rebase Basics
- Pull Requests (PR)
- .gitignore
- README.md Documentation
- Team Collaboration using GitHub

*Learning Approach

To get the most out of this week's learning, focus on understanding how Git tracks changes and how GitHub enables team collaboration. Practice every Git command yourself, experiment with branches, and understand each step of the workflow rather than simply following tutorials.

Keep these points in mind:

- ✓ Practice every Git command on your own repository.
- ✓ Understand the purpose of commits, branches, and merges.
- ✓ Write meaningful commit messages for every change.
- ✓ Keep your GitHub repository organized with a proper README and .gitignore.
- ✓ Verify AI-generated Git commands by testing them locally.

*Complete the following practical exercises:

- Install Git on your system.
- Configure Git with your username and email.
- Create a GitHub account (if you don't have one).
- Create a new GitHub repository.
- Clone the repository to your local machine.
- Add sample project files and commit your changes.
- Push the repository to GitHub.
- Create a new branch and make changes.
- Merge the branch into the main branch.
- Create a professional **README.md** file.
- Configure a **.gitignore** file.
- Prepare a short PDF explaining:
 - Git, GitHub, Git Workflow
 - Branching, Merge
 - Rebase
 - Pull Requests
 - README.md
 - .gitignore

Task Submission

Once you complete the above task, submit:

- ✓ GitHub Repository Link
- ✓ PDF Report (2–4 pages)
- ✓ Screenshot of Git Installation
- ✓ Screenshot of Git Commands
- ✓ Screenshot of Branch Creation & Merge

Task Submission Deadline: Wednesday (EOD)

Scenario

Imagine you have joined a company as a **Junior DevOps Engineer**. Your team is developing a new application, and your responsibility is to manage the project's source code using Git and GitHub while collaborating with other developers.

*Complete the following:

- Create a repository named **DevOps-Week 02**
- Add at least five project files.
- Create two feature branches.
- Make different changes in each branch.
- Merge both branches into the main branch.
- Create a Pull Request (PR).
- Resolve one merge conflict (if possible).
- Update the **README.md** with project details.
- Push the final project to GitHub. .

Activity Submission

After completing the hands-on activity, submit:

- ✓ GitHub Repository Link
- ✓ Screenshot of Branches
- ✓ Screenshot of Pull Request
- ✓ Screenshot of Merge Completion
- ✓ Screenshot of README.md
- ✓ Short implementation summary

Activity Submission Deadline: Friday (EOD)

AI Assistant Task

Use an AI Assistant such as **ChatGPT, GitHub Copilot, Amazon Q, Gemini, Claude, or Microsoft Copilot** to strengthen your understanding of Git and GitHub.

*Complete the following:

- Ask AI to explain the complete Git workflow.
- Generate a Git cheat sheet with **25 commonly used Git commands**.
- Ask AI to explain the difference between **Merge** and **Rebase**.
- Generate a merge conflict scenario and learn how to resolve it.
- Ask AI to explain GitHub Pull Requests and code reviews.
- Verify the AI-generated Git commands by testing them on your local repository.

DevOps Pro Tip

Write clear and meaningful commit messages. A clean Git history makes collaboration easier, simplifies troubleshooting, and improves project management.

Focus on understanding the Git workflow rather than just running commands. Building good version control habits now will make collaboration and CI/CD much easier throughout your DevOps journey.

If you face any genuine issues, feel free to reach out before the submission deadline.